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Recent advances in black box and white-box models for urban heat island prediction: Implications of fusing the two methods

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Article preview
Abstract
Introduction
Section snippets
References (136)

Highlights

- Current use of white-box and black-box modelling of UHI is discussed.
- Future research directions and areas of effective discourse are identified.
- Fusion of white-box and black-box approaches for city design stages are discussed.
- Coupling White-box modelling and machine-learning for city operations are proposed.

Abstract

The urban heat island (UHI) phenomenon is a serious concern for urban planners and policymakers, requiring effective and efficient mitigation policies. To develop such policies, accurate and pre-emptive estimations of current and future UHI manifestations are vital elements that help determine efficient policies and mitigation techniques. There are two fundamental approaches for modelling overheating in an urban environment: white-box and black-box based methods. The first one is characterized by the easily interpretable working process, while the unclear working procedure defines the second one. The present study comprehensively reviews the commonly used white-box and black-box based approaches applied for UHI predictions, analyses the existing literature adopting these tools for UHI prediction, and discusses the effectiveness of fusing both methods at the design and operation stages of the urban area for effective prediction and mitigation of UHI effect. The literature analysis showed that the transparent working process and high prediction accuracy of the physical-based white-box models make them a popular and reliable tool for UHI evaluation. Nevertheless, some white-box based simulation tools are too complex and require a high level of expertise to operate, leading to potential inaccuracies in the obtained outcomes. Black-box models, in turn, despite their opaque working process, are more straightforward in use and require less computation time. The fusion of these two methods is a novel approach that may benefit both UHI prediction and mitigation at the design and operation stages, respectively.

Keywords

UHI; White-box models; Black-box models; Physical-based modelling; Machine learning; The fusion of white box and black box models

Abbreviations

ANN, Artificial Neural Network; ANOVA, Analysis of Variance; ARIMA, Autoregressive Integrated Moving Average model; AWS, Automatic Weather Station; BEM, Building Energy Model; BEP, Building Effect Parameterization; BH, average Building Height; BRT, Boosted Regression Tree; CAT, Canyon Air Temperature model; CFD, Computational Fluid Dynamics; CRENAU, Urban Architecture Nantes Research Centre; CWS, Citizen Weather Station; DEM, Digital Elevation Model; DNN, Deep Neural Network; DT, Decision Tree; EDR, Euclidean distance to roads; EF, Elevation Factor; ETM +, Enhanced Thematic Mapper Plus; EVI, Enhanced Vegetation Index; GNDVI, Green Normalized Difference Vegetation Index; GPI, Gravity Index for Parks; GPS, Global Positioning System; GRU, Gated Recurrent Unit algorithm; GWR, Geographically Weighted Regression; HVAC, Heating Ventilation and Air Conditioning system; KMA, Korea Meteorological Administration; LIME, Local Interpretable Model Agnostic Explanations; LST, Land Surface Temperature; LSTM, Long Short Term Memory algorithm; LULC, Land Use Land Cover; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; MESO-NH, Mesoscale non-hydrostatic model; ML, Machine Learning; MLP-MC, The combination of Multi-Layer Perceptron and Markov Chain; MNDWI, Modified Normalized Difference Water Index; MNUHII, Maximum Night time Urban Heat Island Intensity; NDBI, Normalized Difference Built-up Index; NDBSI, Normalized Difference Bare Soil Index; NDVI, Normalized Difference Vegetation Index; OLI, Operational Land Imager; OLS, Ordinary Least Square regression; PCM, Phase Change Material; PD, Population Density; PID, Proportional Integral Derivative controller; RD, Road Density; RF, Random Forest algorithm; RH, Relative Humidity; RL, Reinforcement Learning; RMSE, Root Mean Square Error; RNN, Recurrent Neural Network; RPSC, Road Pavement Solar Collector system; SH, Sunshine Hours; SLUCM, Single Layer Urban Canopy Model; SURFEX, Surface Externalisée modelling system; SVF, Sky view factor; SINDy, Sparse Identification of Non-linear Dynamics; TEB, Town Energy Balance model; TF, Terrain Factor; TIRS, Thermal Infrared Sensor; TMY, Typical Meteorological Year; UCM, Urban Climate Model; UHI, Urban Heat Island; UHII, Urban Heat Island Intensity; UTFVI, Thermal Field Variance Index; UWG, Urban Weather Generator; WRF, Weather Research and Forecasting model

Previous article in this issue Next article in this issue

Introduction

Urbanization is a crucial element affecting the social and economic development of many countries around the globe. Over the last 50 years, a remarkably swift rise in the rate of urbanization has been observed worldwide. Recent statistics show that in 2020, 56.2% of the global population lived in urban areas [1]. There are predictions that this trend will increase further, and by 2050, about two-thirds of the world's population will live in cities [2]. This rapid urbanization is often accompanied by notably higher urban air temperatures compared to surrounding rural areas. For example, the daily mean difference between urban and rural air temperatures varies from 2 °C to 5 °C, while the maximum difference reaches 12 °C [3]. Such a phenomenon where urban atmospheric and surface temperatures are relatively higher than those in the rural areas or urban peripheries is known as the urban heat island (UHI).

The formation of UHI results from the combined impact of numerous factors inherent to the urbanized areas, such as urban materials with heightened heat absorptive properties, complex urban geometry, excessive anthropogenic heat release, and reduced natural landscapes [4]. For example, most conventional materials used to construct pavements and roofs in urban areas reflect less solar energy and absorb and emit more heat; albedos of urban materials are generally 5–10% lower than in rural areas [5]. Additionally, evapotranspiration processes are often limited due to waterproof materials, further exacerbating heat accumulation in urban areas [6]. For instance, it has been reported that under direct sun exposure, the temperatures of urban surfaces may reach up to 88 °C [6]. In contrast, the moist soil covered with vegetation under the same conditions reaches up to only 18 °C [6].

Furthermore, the dimensions of the buildings and streets impact the wind flow and the buildings' ability to absorb and release heat; mean annual wind speed values in urban areas have been reported to be typically 30–40% lower than those in rural areas [5]. Urban areas are also characterized by relatively higher anthropogenic heat fluxes resulting from high population densities. This higher anthropogenic heat release is a consequence of human activities starting from the human metabolism and finishing with the cooling and heating of buildings, transportation, fuel, and electricity use. The heat produced from these activities modifies the urban thermal structure through various heat transfer processes [6].

The effects of UHI in urban areas are substantial and multi-faceted. Overall, they can be classified under two broad categories: (i) impact on people and (ii) impact on the urban environment. Higher urban temperatures, especially in summertime and extreme heatwave events, notably affect human comfort conditions and health. An empirical example of such an effect is the 2003 European heatwave characterized by temperatures above 40 °C and 70,000 excess deaths [7]. Several studies have established substantial evidence linking urban over warming, heat waves, and the synergetic interactions of the two phenomena to increased mortalities and morbidities, particularly for vulnerable demographics (i.e., the elderly and energy-poor populations) [[7], [8], [9], [10]]. For instance, Huang et al. [7] reported a 28.8% increase in cardiovascular diseases attributed to UHI between 1984 and 2017 in Beijing. UHI has also been reported to influence atmospheric pollution in urban areas. The interactions between UHI and atmospheric air pollutants are well established in the literature. They are primarily due to complex photochemical reactions between UHI-dependent increase in temperature and common air pollutants prevalent in urban areas such as SO_2, NO_2, CO, PM_{10}, and $PM_{2.5}$ [8,9,11]. Moreover, the increase in the UHI intensity (UHII) may cause a noticeable rise in the cooling energy demand, particularly during the summertime, which in turn will contribute to global energy consumption. For example, Su et al. [12] revealed that UHII of 0.5 °C increases cooling energy demand by up to $1.84\,kWh/m^2$ in Seoul. Similarly, Skelhorn et al. [13] reported a 12% increase in cooling energy demand attributed to urban over warming in Manchester, UK.

As discussed above, urban over warming is a severe issue with broad implications on human health and the overall urban ecosystem. It is thus essential to devise effective mitigation measures and policies to counter the effects of increasing urban overheating and its subsequent effects on urban dwellers and the environment. To do so, accurate UHI estimation and forecasting techniques are essential. The existing methods used for estimating and forecasting UHI can be classified into two groups: white-box and black-box methods. In the first case, models with clear inner logic are used to model the complex effect of the UHI on the urban environment. In the second case, machine learning methods, whose working principle is difficult to explain, are adopted. Both approaches allow predicting UHI, mapping it and evaluating various effects contributing to its formation, and analyzing the efficiency of strategies adopted for its mitigation. In the last decades, white-box models have been widely employed by researchers primarily because of their simplicity and transparency. However, the use of white-box models often requires many inputs and a high level of expertise, making them challenging to employ or highly inaccurate. As an alternative, black-box models in UHI studies have increased over the years as they often offer higher estimation/forecasting accuracy than white-box models. While that's the case, black-box methods are highly opaque, making it challenging to determine their inner mechanisms and how they arrive at specific predictions/estimations [14,15].

Despite the importance of accurate UHI predictions/estimations for relevant urban policy and the promotion of the overall urban liveability agenda, the comparative usefulness of white-box and black-box models in predicting/forecasting UHI has not been critically explored and reviewed in the literature. Previous research, particularly review articles on the subject matter, have primarily focused on existing UHI mitigation strategies [[16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]], UHI studies in specific climate [[27], [28], [29], [30]], and the impact of UHI on energy consumption [31]. For example, Jamei et al. [25] reviewed the studies adopting pedestrian level greening and urban geometry modifications to improve outdoor thermal comfort conditions. Brozovsky et al. [28] reviewed publications on UHI research in cold and polar climate regions defined by Koppen-Geiger classification. A few studies are focusing on the approaches used to investigate the UHI. Kim et al. [32] conducted a systematic literature review of the studies to reveal the commonly used methods to estimate UHII. Mirzaei et al. [33], in the review paper, summarized the widely used approaches to study UHI, their benefits, and limitations. The authors classified the existing methods depending on the scale (meso scale, city scale, etc.). They mainly focused on the white-box simulation tools based on physical models and their working principle.

Summing up, the analysis of the existing literature revealed a substantial lack of studies documenting the strategic and differential use of white-box and black-box methods to study UHI behaviour, forming a significant research gap. To address this gap, the present paper reviewed the existing literature on the white-box and black-box methods used to estimate/forecast UHI. The objectives of the study are as follows: (i) introduce the concept of white-box and black-box UHI prediction models as well as their working principle, (ii) summarize their current research status and practical applications, (iii) discuss the fusion of white box and black box models for UHI prediction at the design and operation stages of urban area.

For analysis, the articles from three main databases: Science Direct, Google Scholar, and Scopus, were considered. Additionally, publications from conference proceedings associated with machine learning and building science were involved in the analysis. The following keywords were used during the search process: UHII prediction, urban climate analysis, white-box models, black-box models, machine learning models, and numerical models. The keywords were used as a single item and as a combination of several keywords.

The remaining sections of the paper are organized in the following way. Section 2 summarizes the commonly used white-box models, their working principle, an analysis of studies adopting white-box models for UHI prediction, and a discussion on the highlights and limitations of the white box methods for UHI prediction. Section 3 reviews the black-box models for UHI prediction and their working principle, analyzes studies using them for UHI prediction, and discusses the highlights and limitations of black-box approaches used for UHI prediction. Section 4 discusses fusing white box and black box models for predicting and mitigating UHI at the design and operation stages of the urban area. Section 5 gives conclusive remarks about using the white-box and black-box models to predict UHI.

The present article is helpful to urban climate researchers and policy makers interested in applying white-box and black-box methods in UHI modelling, as well as their working principles, benefits, and limitations. Additionally, the feasibility of fusing white box and black box models for potential improvement of current UHI modelling practices and effective policies was highlighted.

Section snippets

White-box approaches for UHI prediction

White-box models represent the type of models that provide a clear explanation of how they behave, how they produce predictions, and which parameters influence the model. Over the last decades, white-box approaches have been widely used for UHI prediction due to their easily explainable mechanism. Generally, the white box approaches for UHI prediction can be classified into two groups: (i) simulation tools based on physical and mathematical models and (ii) some statistical approaches whose …

Black-box based approaches for UHI prediction

Black-box based methods represent those approaches whose working system consists of inputs and outputs only, without a clear explanation of how it operates. The models can be defined as black-box, whether because the function relating the input and output is too complex to easily comprehend or because the model may be simple in fact, but its properties are not available for analysis. Black-box based approaches do not require the detailed knowledge of on-site physical information of the modelled …

Fusion of white-box and black-box modelling at the urban planning/design stage

A central question revolves around how we can interactively fuse physical-based and black-box modelling techniques to deepen our understanding of UHI, particularly in the early stage of city development projects. Granted that the two approaches can be individually employed to evaluate UHI manifestations, we earlier established some of the inherent fundamental limitations of both white-box models and black box models likely to hinder their deployment in city design stages or in the formation of …

Conclusions

In the present study, the two main approaches used for UHI prediction, namely white-box and black-box methods were discussed. Based on the analysis of the white-box models used to study overheating, it was concluded that despite the transparency of the white-box approaches, there are still some aspects of these methods that remain unclear. For example, the way of incorporating breeze circulations, which considerably contributes to the formation of UHI. Additionally, the literature does not …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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Citation Indexes	58
Captures	
Mendeley Readers	118

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